B13 Interpretability and fairness

Permutation importance, SHAP and subgroup performance.

B13-L1 Permutation importance on held-out data

Measure importance by shuffling on held-out data, with the reported metric, and report the spread.

```
from sklearn.inspection import permutation_importance
result = permutation_importance(model, X_test, y_test, scoring="roc_auc",
                                n_repeats=10, random_state=0)
result.importances_mean, result.importances_std
```

Impurity importance bias	Favours high-cardinality features, from training data. A random ID scored 0.335 in this lesson's example.
Permutation importance	Shuffle a column on held-out data and measure the score drop, with its spread.
Correlated features	Share credit. Permute and report them as a group.
What importance means	What the model relies on. Not what causes the outcome.

B13-L2 SHAP values, and what they do not tell you

SHAP explains one prediction additively in a stated output space. It describes the model, not the world.

```
prediction = base_value + sum of the row's SHAP values
```

```
import shap
explainer = shap.TreeExplainer(model)
explanation = explainer(X_test)          # shap.Explanation
explanation.values, explanation.base_values
```

SHAP additivity	prediction = base_value + sum of the row's SHAP values, in the explainer's output space.
Output space for GradientBoostingClassifier in TreeExplainer	Log-odds (decision_function), not probability.
Global ranking from SHAP	Mean absolute SHAP value per feature.
What SHAP cannot show	Causes, a stable split between correlated features, behaviour outside the training range.

B13-L3 Performance by subgroup, and the fairness trade-offs

Measure rates per group with counts, choose the fairness property that fits the decision, and document the gaps.

Rates to report per group	n, positives, selection rate, TPR, FPR, precision, calibration.
Impossibility result	With different base rates, calibration and equal error rates cannot all hold (Kleinberg et al. 2016; Chouldechova 2017).
Dropping a protected attribute	Proxies keep its influence. Keep it for measurement where lawful.
Choosing a fairness property	From the harm: missed help (TPR), wrong flags (FPR), numbers used directly (calibration).